Single-phase Boost PFC Converter - III

Introduction

In the rapidly evolving landscape of charging infrastructure, efficient and reliable power conversion is paramount. This lecture delves into the intricate details of the single-phase boost Power Factor Correction (PFC) converter, a fundamental building block for AC-to-DC conversion systems that demand high power quality. Building upon previous discussions, we will explore the critical aspects of sizing passive components like inductors and capacitors, and meticulously define the voltage and current ratings for the active switching devices and diodes. Understanding these design considerations is crucial for engineers to develop robust and optimized power converters that meet stringent performance requirements, ensuring stable and efficient energy delivery for various applications, including electric vehicle charging.

Core Concepts

Term	Definition	Significance
`Unity Power Factor`	A condition where the current drawn from the AC source is perfectly in phase with the voltage, minimizing reactive power and maximizing real power transfer.	Essential for efficient power utilization and compliance with grid regulations, reducing losses and improving system performance.
`Second-Line Harmonic`	A ripple component in the output voltage or current that occurs at twice the input line frequency (e.g., 100 Hz for a 50 Hz input).	This harmonic significantly impacts the sizing of the output capacitor, often leading to large component dimensions in single-phase converters.
`Duty Ratio (D)`	The fraction of a switching period during which a power switch is in its ON state. In PFC, it varies to maintain a constant output voltage.	Crucial for controlling the converter's output voltage and shaping the input current to achieve power factor correction.

Term	Definition	Significance
`RMS Current`	The root mean square value of a time-varying current, representing its heating effect equivalent to a DC current.	Fundamental for selecting appropriate power switches and diodes, as it dictates their thermal stress and power dissipation capabilities.
`Voltage Ripple (ΔV_o)`	The small, undesirable AC variation superimposed on the DC output voltage of a converter.	A key performance metric for power supplies; minimizing it ensures stable operation of the connected load and influences capacitor sizing.
`V_o > V_s,peak`	The condition that the output DC voltage of a boost converter must be greater than the peak value of the input AC voltage.	A fundamental operational constraint for boost converters, ensuring proper voltage step-up and control authority.
`Allowable Ripple Current`	The maximum permissible AC component of current that an inductor can handle without excessive heating or saturation.	Directly influences the required inductance value; a smaller ripple allows for a smaller inductor but may require higher switching frequency.
`Power Balance`	The principle that, assuming negligible losses, the instantaneous input power to a converter equals its instantaneous output power.	Used to derive relationships between input and output quantities, particularly for calculating capacitor current and ripple.

Detailed Analysis

1. Recap of Single-Phase Boost PFC Converter Operation

The single-phase boost PFC converter typically consists of a full-bridge diode rectifier followed by a boost converter stage. The rectifier converts the AC input voltage ($V_s = V_{s,pk} \sin(\omega t)$) into a rectified DC voltage (V_d), which is the modulus of the input AC voltage. This V_d then serves as the input to the boost converter.

Key operational aspects previously discussed include:
* **High-Frequency Switching**: The switches (S1 and S2) operate at a frequency (f) significantly higher than the AC supply frequency (f_s), ensuring that the rectified DC voltage appears constant over a switching period.
* **Varying Duty Ratio**: To maintain a constant output voltage (V_o) across the load, the duty ratio (D) of switch S1 must vary dynamically. This duty ratio ranges between 0 and 1, with its specific value dependent on the instantaneous input voltage and the desired output voltage.
* **Output Voltage Constraint**: A critical condition for boost converter operation is that the output voltage (V_o) must always be greater than the peak input voltage ($V_{s,pk}$). For instance, if the input is 230V RMS ($V_{s,pk} \approx 325V$), the output V_o typically needs to be 400V or higher.
* **Inductance Sizing**: The required inductance value (L) depends on the output voltage (V_o), the maximum allowable ripple current (typically 10% of $I_{s,pk}$), and the switching frequency (f). Higher switching frequencies allow for smaller inductance values.

2. Capacitance Sizing for Output Voltage Ripple

Sizing the output capacitor (C) is crucial for maintaining the output voltage within an allowable ripple. The design process for the capacitor relies on several key assumptions:
* **Unity Power Factor (UPF)**: The converter is assumed to draw current from the grid in phase with the voltage, meaning $I_s(t) = I_{s,pk} \sin(\omega t)$ and $V_s(t) = V_{s,pk} \sin(\omega t)$.
* **Negligible Losses**: For initial calculations, losses in the rectifier and boost converter are assumed to be negligible, allowing for a power balance between input and output.

The input power (P_{in}) can be expressed as:
$P_{in}(t) = V_s(t) * I_s(t) = (V_{s,pk} \sin(\omega t)) * (I_{s,pk} \sin(\omega t)) = V_{s,pk} I_{s,pk} \sin^2(\omega t)$

Using the trigonometric identity $\sin^2(\theta) = (1 - \cos(2\theta))/2$, this becomes:
$P_{in}(t) = (V_{s,pk} I_{s,pk} / 2) * (1 - \cos(2\omega t))$
$P_{in}(t) = (V_{s,pk} I_{s,pk} / 2) - (V_{s,pk} I_{s,pk} / 2) \cos(2\omega t)$

By applying power balance ($P_{in}(t) = P_D(t)$), where $P_D(t)$ is the power delivered to the output stage (parallel combination of C and R_L), and $P_D(t) = I_{conv}(t) * V_o$, we can find the converter current ($I_{conv}(t)$):
$I_{conv}(t) * V_o = (V_{s,pk} I_{s,pk} / 2) - (V_{s,pk} I_{s,pk} / 2) \cos(2\omega t)$
$I_{conv}(t) = (V_{s,pk} I_{s,pk} / (2V_o)) - (V_{s,pk} I_{s,pk} / (2V_o)) \cos(2\omega t)$

This $I_{conv}(t)$ comprises two parts:
* A DC component: $I_L = V_{s,pk} I_{s,pk} / (2V_o)$, which flows into the load R_L.

* A time-varying component: $I_c(t) = - (V_{s,pk} I_{s,pk} / (2V_o)) \cos(2\omega t)$, which flows into the capacitor C. This component is a **second-line harmonic current**, varying at twice the input line frequency (e.g., 100 Hz for 50 Hz input).

The voltage ripple (V_{o_ripple}) across the capacitor due to this current is calculated by integrating the capacitor current:
$V_{o_ripple}(t) = (1/C) \int I_c(t) dt$
$V_{o_ripple}(t) = (1/C) \int - (V_{s,pk} I_{s,pk} / (2V_o)) \cos(2\omega t) dt$
$V_{o_ripple}(t) = - (V_{s,pk} I_{s,pk} / (2V_o * 2\omega C)) \sin(2\omega t)$

The peak-to-peak ripple voltage (ΔV_o) is twice the peak ripple:
$\Delta V_o = 2 * |V_{o_ripple,peak}| = (V_{s,pk} I_{s,pk} / (2V_o * \omega C))$

Substituting $I_L = V_{s,pk} I_{s,pk} / (2V_o)$, we get:
$\Delta V_o = I_L / (\omega C)$

Rearranging for C:
$C = I_L / (\omega * \Delta V_o)$

Since $I_L = P_L / V_o$ (where P_L is the load power) and $\omega = 2\pi f_s$ (where f_s is the line frequency):
$C = P_L / (2\pi f_s * \Delta V_o * V_o)$

This formula highlights that the capacitor size is inversely proportional to the line frequency and the allowable voltage ripple. The presence of the second-line harmonic frequency ($2f_s$) means that for typical line frequencies (50/60 Hz), the ripple frequency (100/120 Hz) is relatively low. This low frequency necessitates a significantly larger capacitance value compared to converters operating at higher switching frequencies, making the output capacitor a major bottleneck in terms of size and cost for single-phase AC-to-DC converters.

3. Switch S1 Ratings: Voltage and RMS Current

Defining the ratings for the power switches (S1 and S2) is critical for component selection and reliable operation.

Voltage Rating for S1:
When switch S2 is ON, the entire output voltage (V_o) appears across switch S1. Therefore, the voltage rating for S1 (V_{S1}) must be at least equal to V_o.
$V_{S1} = V_o$
Practically, a safety margin of 40% to 60% is typically applied. For example, if V_o is 400V, a switch rated for 600V to 700V would be selected (e.g., 1.4 * 400V = 560V, so 600V or 650V is common). This margin accounts for voltage spikes and ensures robust operation.

RMS Current Rating for S1:
To determine the RMS current (I_{S1_RMS}) through switch S1, we first establish a

general method for calculating RMS current in a generic power converter switch. For a switch (Sa) in a half-bridge that conducts a constant current (I_o) for a fraction (D) of the switching period (T_s), the RMS current is:
$I_{RMS_Sa} = I_o * \sqrt{D}$

Applying this concept to switch S1 in the boost PFC converter:
* The current flowing through S1 when it is ON is the inductor current, which is approximately equal to the rectified input current (I_d) in the positive half-cycle.
* I_d is approximately $I_{s,pk} \sin(\omega t)$ during the positive half-cycle (0 to T/2).
* The duty ratio (D(t)) for S1 varies with time.

The RMS current for S1 is calculated by integrating the square of the current over a half-cycle and taking the square root of the average:
$I_{S1_RMS} = \sqrt{[(1 / (T/2)) \int_0^{(T/2)} (I_s(t) * D(t))^2 dt]}$
Where $I_s(t) = I_{s,pk} \sin(\omega t)$ and $D(t) = 1 - (|V_s(t)| / V_o) = 1 - (V_{s,pk} \sin(\omega t) / V_o)$ for the positive half-cycle.

This integral is complex and typically evaluated using high-end mathematical software. To perform this calculation, several parameters must be known:
* **$V_{s,pk}$**: Peak input AC voltage (derived from RMS input voltage, e.g., $\sqrt{2} * 230V$).
* **V_o**: Desired output DC voltage.
* **$I_{s,pk}$**: Peak input current. This can be estimated from the load power (P_L) and the input voltage. Assuming 100% efficiency, $P_{in} = P_L$. For a unity power factor, $P_{in} = (V_{s,pk} * I_{s,pk}) / 2$. Thus, $I_{s,pk} = (2 * P_L) / V_{s,pk}$.

Once I_{S1_RMS} is calculated, appropriate switches (e.g., MOSFETs or IGBTs) can be selected. For MOSFETs, the RMS current rating is often sufficient. For IGBTs, both RMS and average current ratings are important for proper selection.

Key Takeaways

- [ ] The output voltage of a single-phase boost PFC converter must always be greater than the peak input AC voltage.
- [ ] The duty ratio of the boost switch varies dynamically to maintain a constant output voltage and achieve unity power factor.
- [ ] The output capacitor in a single-phase PFC converter experiences a second-line harmonic current ripple, leading to a voltage ripple at twice the line frequency.
- [ ] The low frequency of the second-line harmonic ripple necessitates a large output capacitance, which is a significant design challenge for single-phase AC-to-DC converters.
- [ ] The voltage rating of switch S1 must be equal to or greater than the output DC voltage (V_o), with an added safety margin.

- [ ] RMS current ratings for switches are crucial for thermal design and component selection, calculated by integrating the square of the current over a switching cycle.
- [ ] The peak input current ($I_{s,pk}$) can be estimated from the converter's power level and input voltage, assuming unity power factor.
- [ ] High-end mathematical software is often required to accurately calculate the RMS current through the main boost switch (S1) due to the time-varying duty ratio and input current.

Conclusion

The detailed analysis of component sizing and switch ratings for the single-phase boost PFC converter underscores the complexities involved in designing high-performance power electronics. From understanding the impact of second-line harmonics on capacitor size to meticulously calculating RMS currents for active switches, each step is critical for ensuring the converter's efficiency, reliability, and longevity. The insights gained from this lecture are directly applicable to the development of advanced charging infrastructure, where robust and compact power conversion solutions are in high demand. Future discussions will extend to the ratings of other components like switch S2 and the diodes, and delve into the implementation of closed-loop control, further refining our ability to engineer sophisticated power systems.